

STATISTICAL ANALYSIS: INFERENCE & PREDICTION

COMM50 Postgraduate Study & Research Methods

Reminder: Why do experiments?

- At many points of the research and development cycle you will want to demonstrate that your theory/idea offers something new
 - ▣ Or an improvement upon what already exists
- To do this, you will have to carry out some kind of research
- Could be consumer or market research or could be a test of the efficacy of a technology or innovation
- In the case of the latter, an experimental approach is more likely to be applicable

Fairy lasts 2x longer
than the next best-selling brand



Over the years, generations of families have trusted Fairy's cleaning power because it lasts a long, long time. And today it lasts twice as long as the next best-selling brand.

That's **FAIRYCONOMY**



Reduces 90% of plaque germs
that cause gum problems*

*vs. ordinary, non-antibacterial toothpaste



“eight out of ten owners said their cat prefers it”

Hand sanitizers kill
99.99% of germs*



Bias Towards Positive Evidence



“It is the peculiar and perpetual error of the human understanding to be more moved and excited by affirmatives than negatives.”

(Francis Bacon)

- We tend to favour positive evidence (that supports our intuition) more than negative evidence
- Remember: even ‘unwanted’ research outcomes are still valid research!

Randomness

- The best odds in Las Vegas are in the game of roulette
 - ▣ Bet on RED or BLACK outcome
 - ▣ \$1 bet = \$2 return, if successful
- 38 possible outcomes
 - ▣ Numbers 1 to 36, coloured RED & BLACK
 - ▣ Numbers 0 and 00, coloured GREEN



Randomness

- *What's happening here?*



‘Making Sense’ or Randomness

- Sometimes, things just happen by chance
- As humans, we often try to make something out of nothing
 - ▣ Shapes in clouds
 - ▣ Man in the moon
 - ▣ Hidden messages in records played backwards
- The human ability to spot patterns helps us make sense of the world around us & we are creatures of habit
- But, we are sometimes too eager to attribute causality

‘Making Sense’ or Randomness

- The problem with “Winning Streaks” in games of chance is an example
- Thomas Gilovich’s experiment
- Imagine you bet on Red (**R**), where are the “winning runs”?
- **B R R R B R R R B R R B B B R B B R R B B**
 - ▣ 6 of the first 8 were wins
 - ▣ Oh, actually, 8 of the first 11 were wins
 - ▣ This can’t be random!
- No! These are randomly generated outcomes!

‘Making Sense’ or Randomness

- The experiment shows that we are not too good at distinguishing a real pattern from background noise
- This is why we always use statistics and not only intuition in research (this is a science, remember!)
- Further supporting the need for objective hypothesis testing
- We deduce therefore, some ‘failings’ of humans:
 - ▣ We [humans] see patterns where there is only random noise
 - ▣ We see causal relationships where there are none

Inferential Statistics

- Statistical analysis where we want to make decisions
- Often, we deal with *independent* and *dependant* variables
- Determine if two sample data sets are drawn from the same population
- Provides an account of *randomness*
- In effect, this provides a way to indicate whether an intervention has had a significant impact
- Many tests, depending upon data type, scenario

Inferential Statistics

- Deal with statistical analysis where we want to make decisions or build models of a situation based on gathered data
- Typically, we look for certain things
 - ▣ *Is there a relationship between variables?*
 - ▣ *Is something better than an alternative?*
 - ▣ *Is one way of doing something different to another?*
 - ▣ *What might happen if...?*
- The use of statistics removes human subjectivity and the potential for bias / placebo effect

Inferential Statistics

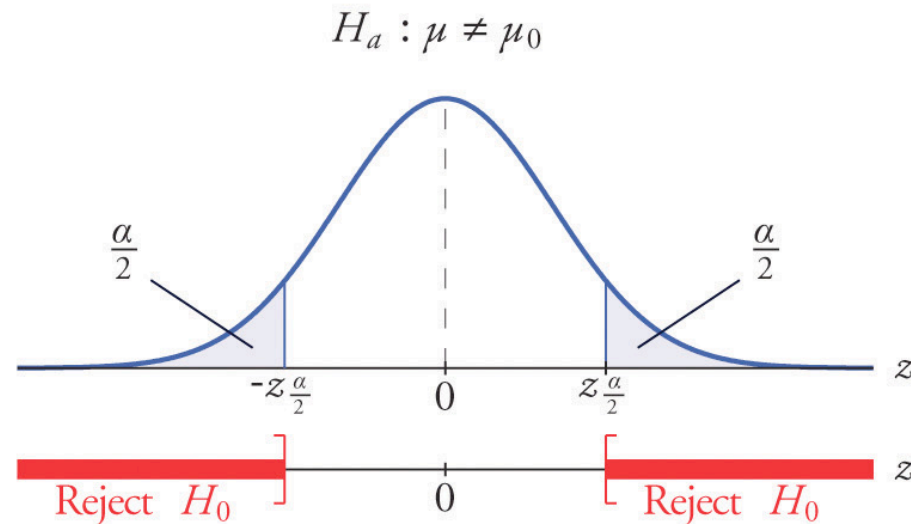
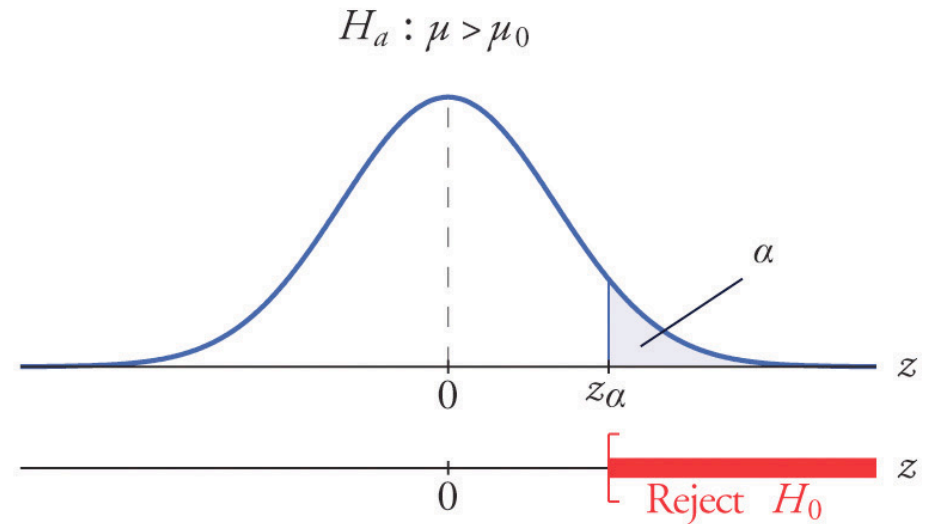
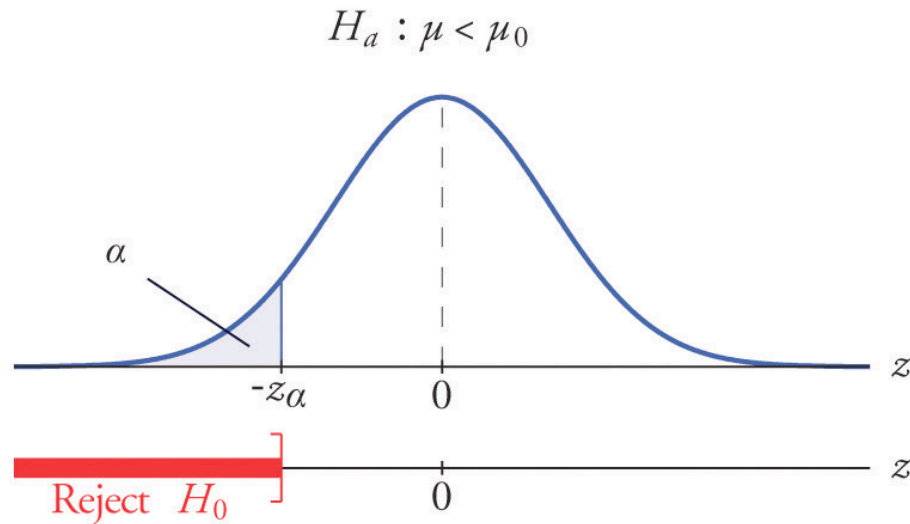
Data Type	Two Group Significance Test	Multiple Group Significance Test	Correlation
Nominal	Chi-Square χ^2	Contingency Table χ^2	Pearson ϕ
Ordinal	Mann-Whitney U	Kruskal-Wallis	Spearman ρ
Interval / Ratio	Student's t-Test	ANOVA	Pearson r

Statistical Tests

- One-tailed tests seeks directional difference between the two sets (i.e. larger or smaller)
 - ▣ Alpha value appears at one 'end' of the distribution
 - ▣ Sensitivity to significance is greater
- Two-tailed tests seeks a difference existing, regardless of direction
 - ▣ Alpha value is split ($\alpha/2$) over both ends of the distribution
 - ▣ Sensitivity to significance is decreased

Statistical Tests: Tails

Image from: https://catalog.flatworldknowledge.com/bookhub/reader/3318?e=fwk-shafer-ch08_s02



Statistical Tests



- Unpaired / Paired
 - ▣ Unpaired assumes samples not related
 - ▣ Paired tests assume the same object is measured under both conditions (e.g. user A before and after)
- Type I error / false positive
 - ▣ Rejecting the null hypothesis when it is true
- Type II error / false negative
 - ▣ Maintaining the null hypothesis when it is false

Chi-Squared Test

- Pearson's Chi-Squared test compares our observed (O) experimental results to what we would expect (E), based on chance

- Chi-Squared formula:

$$\chi^2 = \sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$$

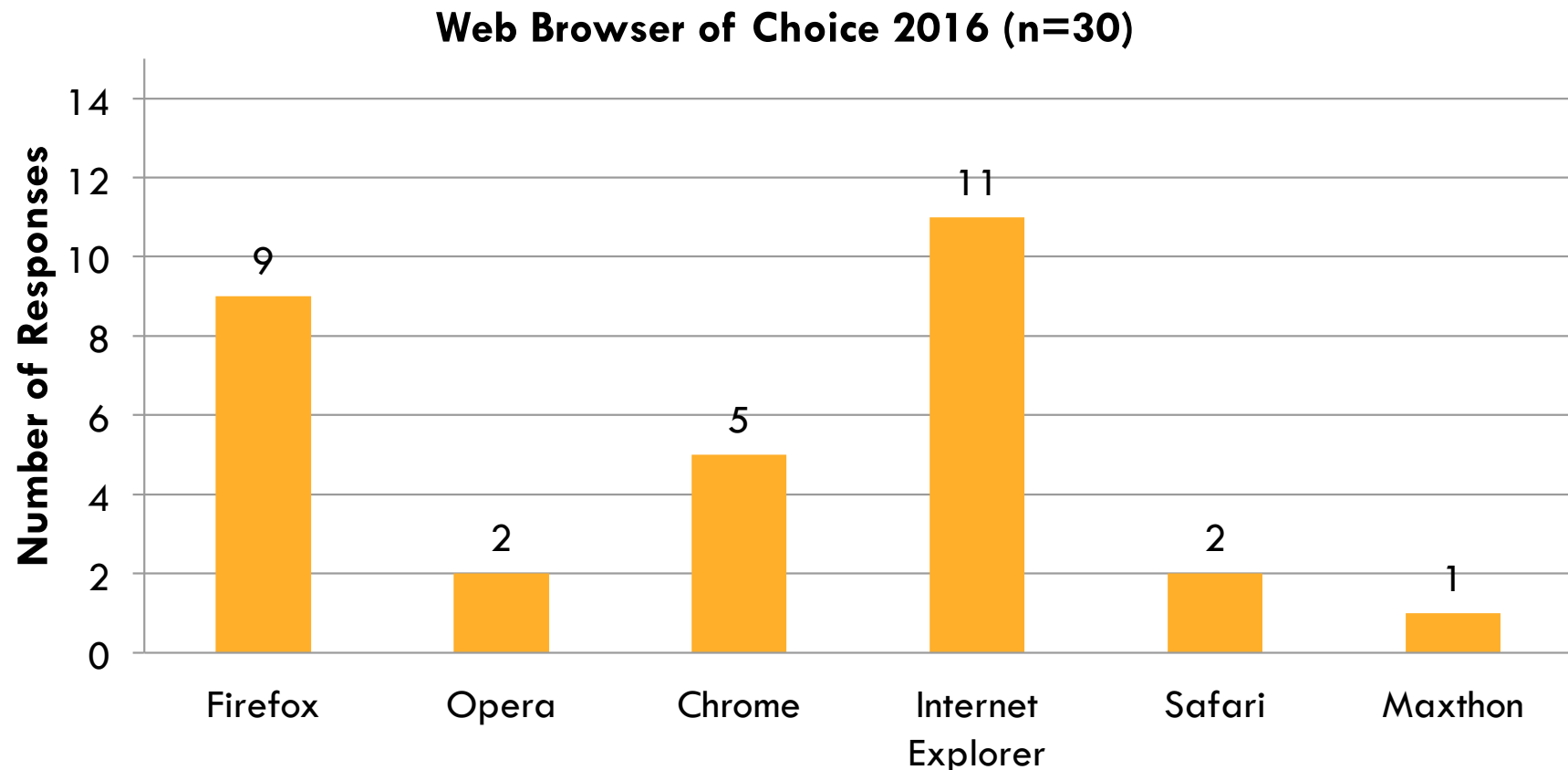
- We employ a table of critical values to compare the χ^2 statistic against (or software can do this for us)
- If χ^2 is greater than the critical value, the results are significant

Chi-Squared Example

- Suppose we are interested in what browser general desktop computer users pick
- We have the idea that the favourite changes frequently
- Null Hypothesis (H_0)
 - ▣ *General desktop computer users to not change their browser of choice frequently*
- Alternate Hypothesis (H_1)
 - ▣ *General desktop users do change their browser of choice frequently*
- To investigate, we use two surveys, a year apart

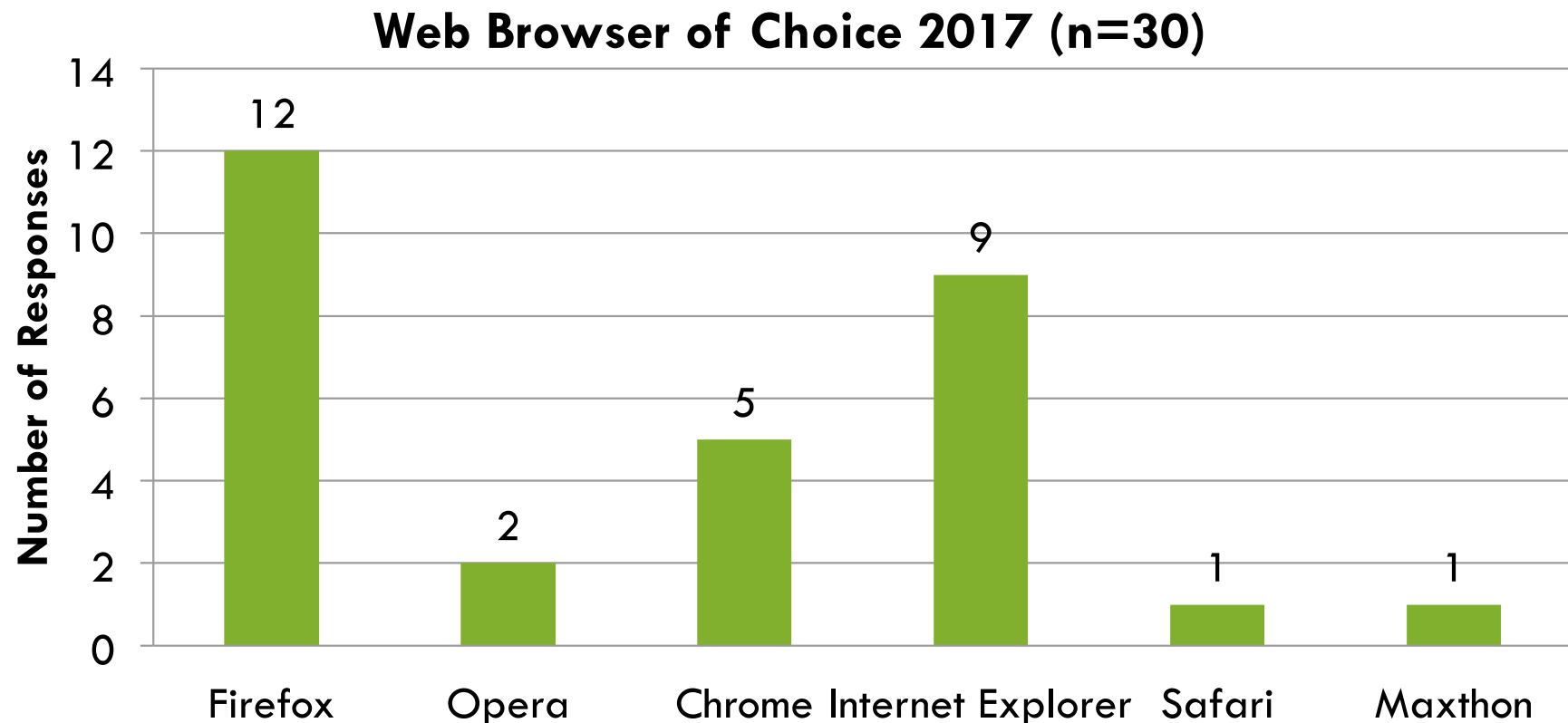
Chi-Squared Example

- In December 2016, we asked a user group (n=30) what their web browser of choice is



Chi-Squared Example

- In December 2017, we asked another user group (n=30) what their web browser of choice is



Chi-Squared Example

- We can use the Chi-Squared test to determine if there is a significant difference between the results

Browser	Expected (2016)	Observed (2017)	$(O-E)^2$	$(O-E)^2/E$
Firefox	9	12	9	1.00
Opera	2	2	0	0.00
Chrome	5	6	1	0.20
Internet Explorer	11	9	4	0.36
Safari	2	1	1	0.50
Maxthon	1	0	1	1.00
$N =$	30		$\chi^2 =$	3.06

Chi-Squared Example

- $\chi^2 = 3.06$
- $df = 5$ (6 categories - 1)
- Working at 95% confidence
- $\chi^2 < 11.07$
- So, the results we see here are not significant
- We maintain the null hypothesis
 - ▣ Something else responsible for the result

df	P = 0.05	P = 0.01	P = 0.001
1	3.84	6.64	10.83
2	5.99	9.21	13.82
3	7.82	11.35	16.27
4	9.49	13.28	18.47
5	11.07	15.09	20.52
6	12.59	16.81	22.46
7	14.07	18.48	24.32
8	15.51	20.09	26.13
9	16.92	21.67	27.88
10	18.31	23.21	29.59

Chi-Squared: Contingency Tables

- Often, categorisation is done across multiple dimensions (variables)
- We may be interested in determining if the one of the variables is contingent upon another
 - ▣ Or, are the categories independent of one another?
- For example: say we are interested in examining first-time level completion rates, of a demo game, in male and female learner players
 - ▣ We survey the number of completion results during a large scale trade expo over the duration of a week

Contingency Table Example

- From surveying 358 gamers, we find the following observed data:

Gender	Completed	Did Not Complete
Female	153	24
Male	105	76

- From an initial inspection, we might hypothesise that female gamers are more likely to complete the level (86%; 153 of 177) than males (58%; 105 of 181)

Contingency Table Example

- We build a Contingency Table to analyse further
- Contingency tables express the expected values, assuming the two variables (gender and pass/fail) are independent
- For each data cell in the table, the expected frequency E_{ij} is calculated:

$$E_{ij} = \frac{R_i C_j}{N}$$

R_i (row total)

C_j (column total)

N (total observations)

Contingency Table Example

- We insert Row and Column totals:

Gender	Completed	Did Not Complete	Total
Female	153	24	177
Male	105	76	181
Total	258	100	358

- Then expected values (in parentheses):

Gender	Completed	Did Not Complete	Total
Female	153 (127.559)	24 (49.441)	177
Male	105 (130.441)	76 (50.559)	181
Total	258	100	358

Remember expected values are $(\text{RowTotal} * \text{ColumnTotal} / \text{TotalObservations})$

Contingency Table Example

- Now we have expected values, we calculate χ^2 using each data cell in the table

Gender	Completed	Did Not Complete	Total
Female	153 (127.559)	24 (49.441)	177
Male	105 (130.441)	76 (50.559)	181
Total	258	100	358

$$\chi^2 = \frac{(153 - 127.559)^2}{127.559} + \frac{(24 - 49.441)^2}{49.441} + \frac{(105 - 130.441)^2}{130.441} + \frac{(76 - 50.559)^2}{50.559}$$

$$= 35.93$$

Contingency Table Example

- $\chi^2 = 35.93$
- Degrees of freedom for contingency tables is:
 - ▣ $df = (R-1)*(C-1)$
 - ▣ R =total rows; C =total columns
 - ▣ So, in this example, $df = 1$
- $\chi^2 > 3.84$
- We reject the null hypothesis:
 - ▣ There is a significant relationship between gender and level completion rate

df	P = 0.05	P = 0.01	P = 0.001
1	3.84	6.64	10.83
2	5.99	9.21	13.82
3	7.82	11.35	16.27
4	9.49	13.28	18.47
5	11.07	15.09	20.52
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Student's t-Test



- Devised by William S Gosset ('Student')
- Parametric: interval and ratio data
- Analysis of the ratio between means and variance of data sets
- If $t > \text{critical value}$, result is significant

$$t = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}}$$

1. http://en.wikipedia.org/wiki/William_Sealy_Gosset
2. <http://www.binbin.net/photos/guinness/Gui/Guinness-Label-Pint-Glasses-Pack-of-2.jpg>

Table from: http://en.wikipedia.org/wiki/Student%27s_t-distribution

<i>One Sided</i>	75%	80%	85%	90%	95%	97.5%	99%	99.5%	99.75%	99.9%	99.95%
<i>Two Sided</i>	50%	60%	70%	80%	90%	95%	98%	99%	99.5%	99.8%	99.9%
1	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	127.3	318.3	636.6
2	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	14.09	22.33	31.60
3	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	7.453	10.21	12.92
4	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	5.598	7.173	8.610
5	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	4.773	5.893	6.869
6	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	4.317	5.208	5.959
7	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.029	4.785	5.408
8	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	3.833	4.501	5.041
9	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	3.690	4.297	4.781
10	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	3.581	4.144	4.587
11	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	3.497	4.025	4.437
12	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.428	3.930	4.318
13	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.372	3.852	4.221
14	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.326	3.787	4.140
15	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.286	3.733	4.073
16	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.252	3.686	4.015
17	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.222	3.646	3.965
18	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.197	3.610	3.922
19	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.174	3.579	3.883
20	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.153	3.552	3.850

$$df = (n_x + n_y) - 2$$

Student's t-Test: Critical Values

Student's t-Test Example

- Observation (not mine)
- “*Chocolate intake is associated with better cognitive function*” (Crichton et al., 2016)
- Alternative Hypothesis
 - ▣ Eating chocolate will improve general knowledge test performance in students
- Null Hypothesis
 - ▣ Eating chocolate will produce no difference in general knowledge test performance in students

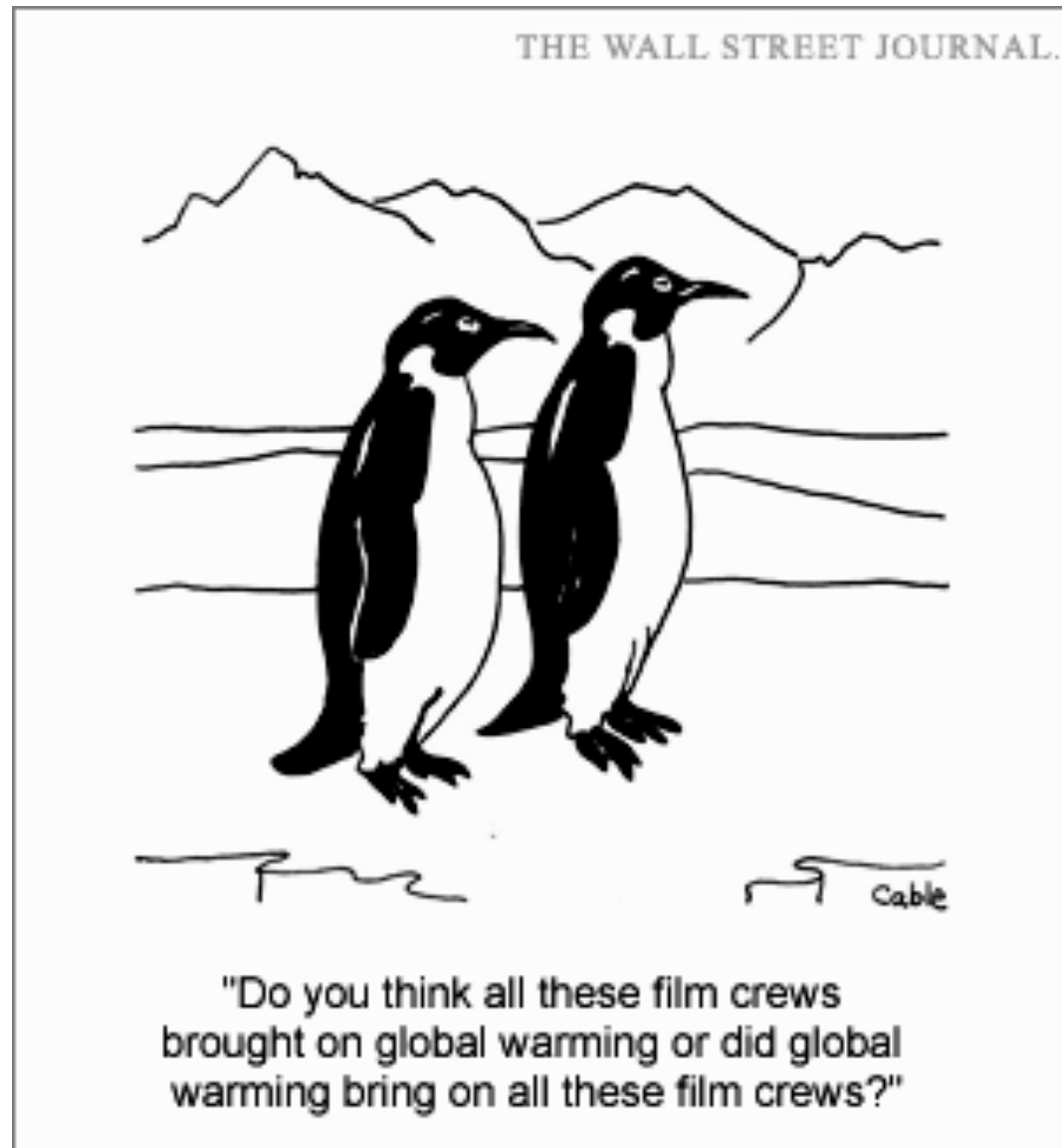
Crichton, G. E., Elias, M. F., & Alkerwi, A. A. (2016). Chocolate intake is associated with better cognitive function: The Maine-Syracuse Longitudinal Study. *Appetite*, 100, 126-132.

Student's t-Test Example

- Randomly select 100 people from population and allocate them to either a control or test group
- Both groups administered a general knowledge quiz (scored out of 30)
 - ▣ $t = -0.37$ [Two-tail crit value = 2.01 ($p=0.713$)]
- $t < 2.01$: the result is not significant: maintain null

	Control Group	Test Group
Mean	15.16	16.04
Variance	69.97	71.54
St. Dev	8.37	8.46

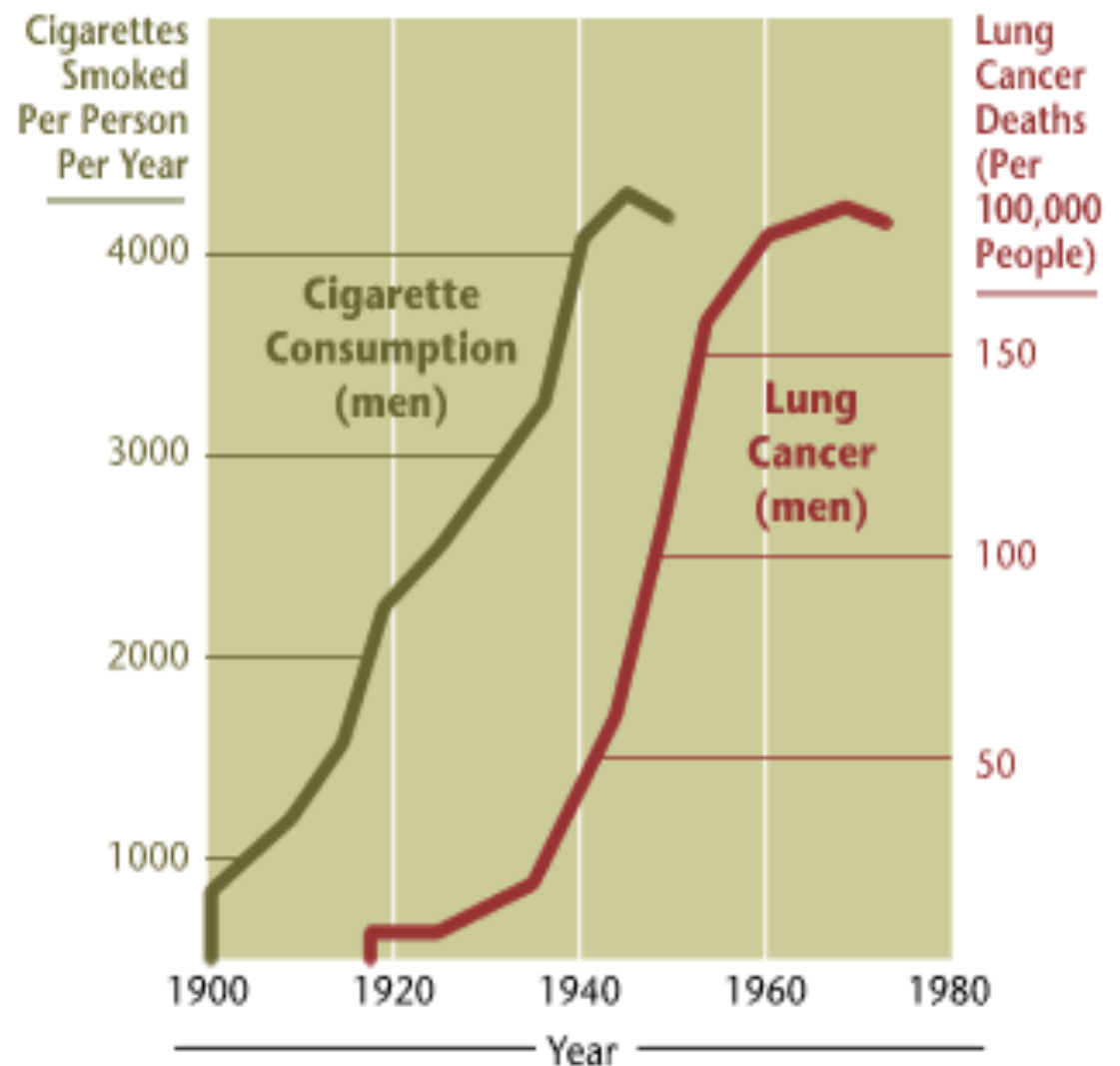
Correlation



Correlation

- Determining if a relationship exists between multiple variables
- Correlation ranges
 - $r(X,Y)=0$: there is definitely no correlation;
 - $r(X,Y)= \pm 1$: there absolutely is a correlation;

20-Year Lag Time Between Smoking and Lung Cancer



Correlation and Regression

- Correlation describes whether variables which describe a subject are related
 - ▣ e.g. *does render time increase with number of voxels?*
 - ▣ e.g. *does download speed correlate with the number of servers in a cluster?*
 - ▣ e.g. *does user acceptance of a robot decrease as its realness increases?*
- Regression is the determination of a mathematical expression of the correlation
 - ▣ For example, linear: $y = ax + b$

Correlation

- Pearson Correlation (interval/ratio):

$$r(X, Y) = \frac{\text{cov}(X, Y)}{\sigma_X \sigma_Y}$$

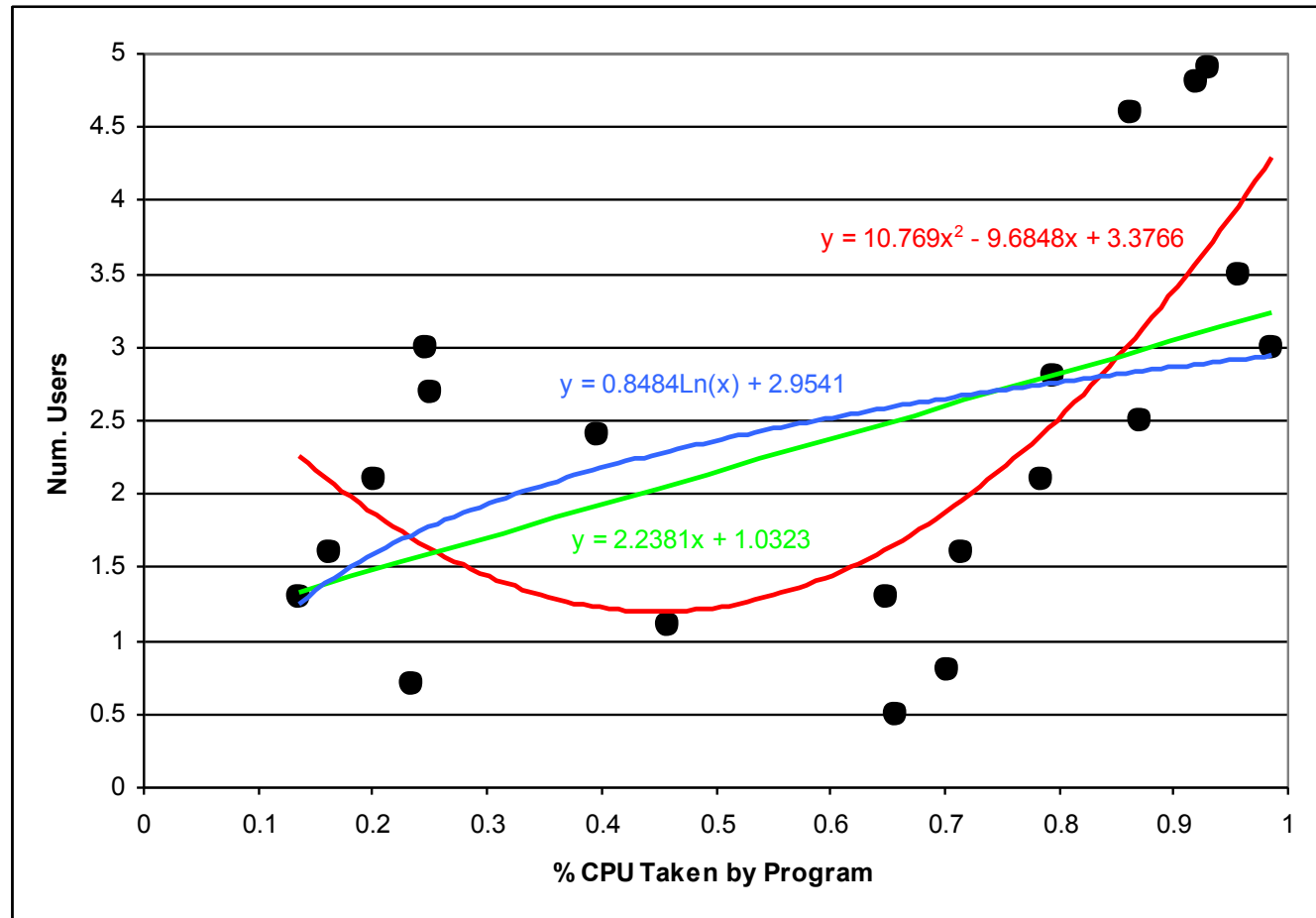
- Where co-variance is:

$$\text{cov}(X, Y) = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{N}$$

- Spearman Correlation (nominal): take Pearson of the ranks of values, rather than values themselves

Correlation and Regression Example

- Number of users of a program vs. % CPU time used on server ($r = 0.51, p < 0.05$)



Correlation and Causality



[Correlation does NOT imply causation!](#) (0 – 2m 45s)

...but causation can be investigated separately

Applied Experimental Design and Statistics

- A brief summary and some pragmatic issues:



- “*Battling Bad Science*” – Ben Goldacre

http://www.ted.com/talks/ben_goldacre_battling_bad_science.html



References



- Social Research Methods: The t-Test
 - http://www.socialresearchmethods.net/kb/stat_t.php
- Rice University: Student's t-Test
 - <http://www.ruf.rice.edu/~bioslabs/tools/stats/ttest.html>